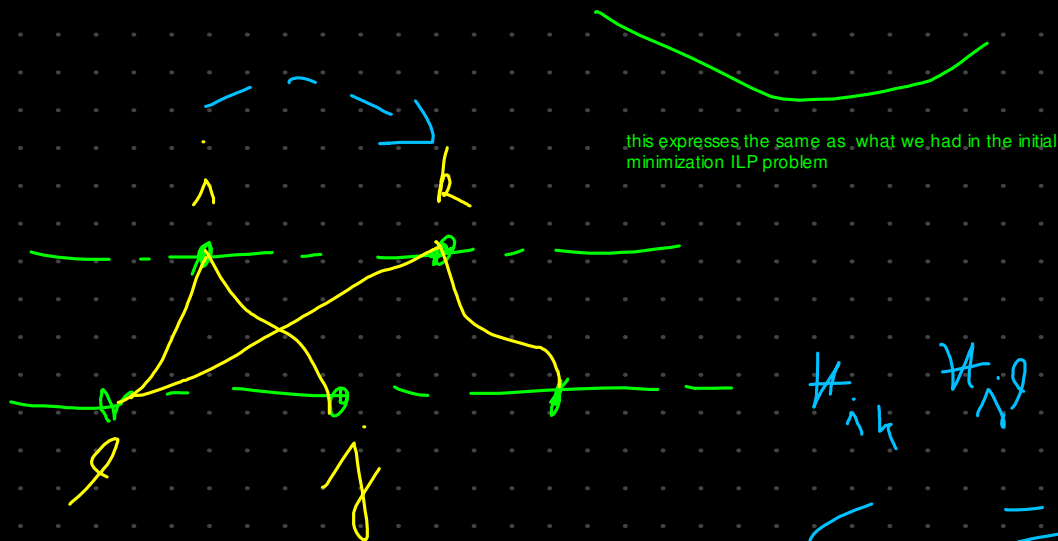


$$x_{ij}^n = 1 - x_{ij}^{n-1} \quad (3) \quad / \quad C_{ijkl}^n = \begin{cases} 1 & (i,j) \& (k,l) \text{ cross} \\ 0 & \text{otherwise} \end{cases}$$



We just need to do this for each edge combo ? and for each level

$$\sum_{\lambda=1}^P \sum_{\substack{(i,j) \\ (k,l)}} C_{ijkl}^{\lambda}$$

$$E_{\lambda} \subseteq V_n \times V_{n+1}$$

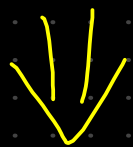
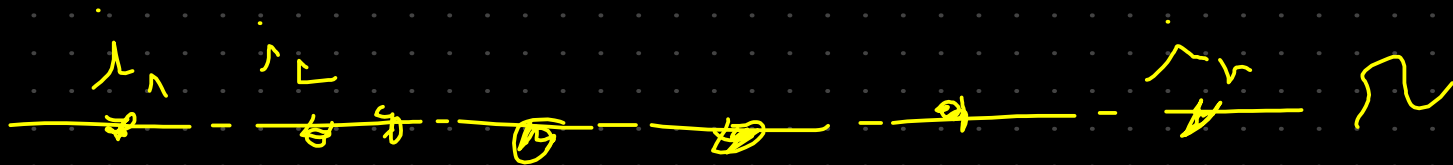
Vertex-exchange graph

Level graph $G = (V, E)$

vertex-exchange graph of G :

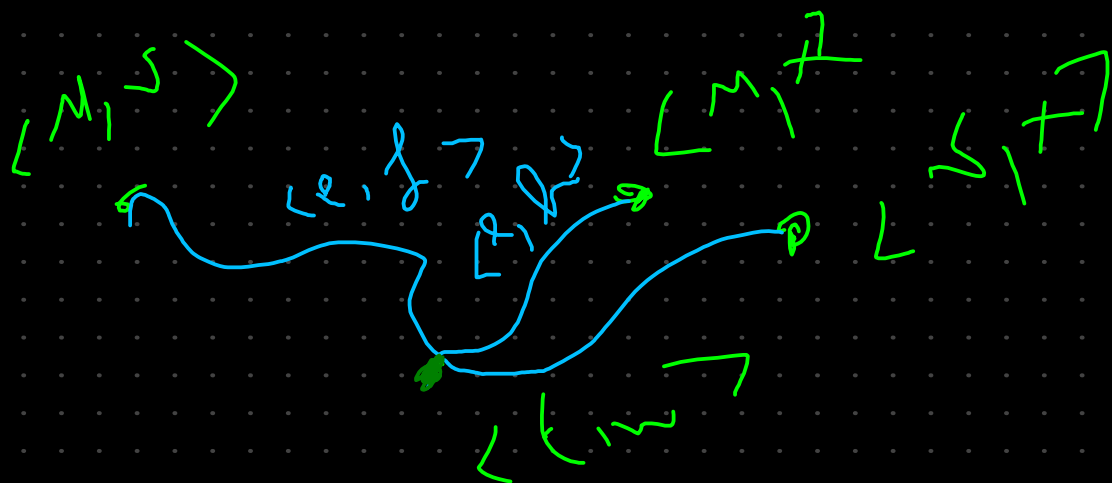
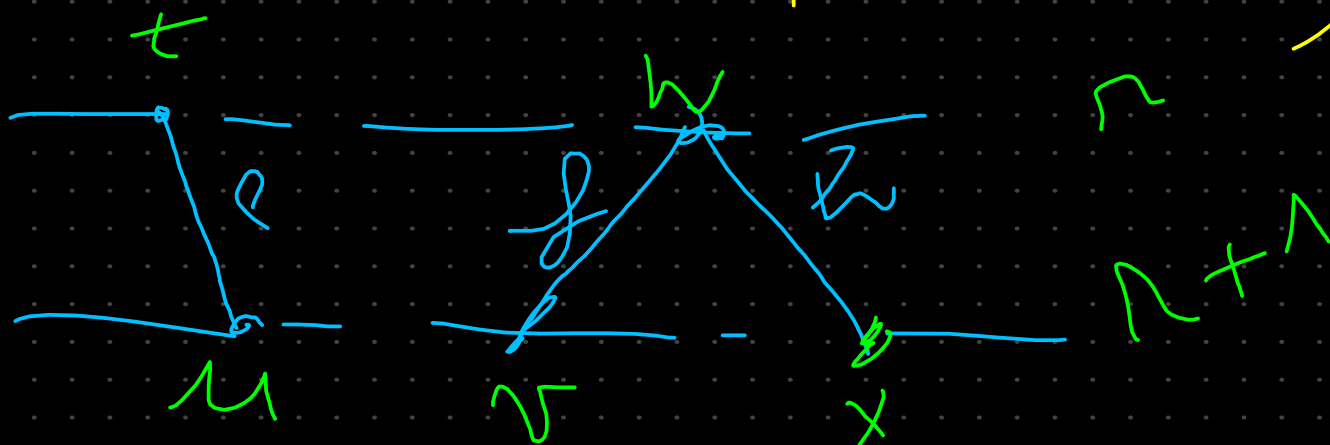
$$\mathcal{G} = (V, \mathcal{E}) \quad / \quad V = V_0 \cup V_1 \cup \dots \cup V_P$$

where $V_n = \{ \langle v, w \rangle \mid v, w \in V_n \}$



$$V_n = \{ \langle i_1, i_2 \rangle, \dots, \langle i_{n-1}, i_n \rangle \}$$

$$E_n = \left\{ \langle e, f \rangle \mid e, f \in E_n, e = (t, u), f = (w, v), \langle t, w \rangle \in V_n, \langle u, v \rangle \in V_{n+1} \right\}$$



Properties of the vertex-exchange graph

jump this section for now

To Do: Level graph planarity and layout